

Auteur: Abdul Ahmed
 Studierichting: Informatica
 Oefeningenreeks 3A nr. 22.9

$$\begin{aligned}
 & \frac{d}{dx} \left(\sqrt{\frac{x-1}{x+1}} \right) \\
 &= \frac{1}{2\sqrt{\frac{x-1}{x+1}}} \cdot \frac{d}{dx} \left(\frac{x-1}{x+1} \right) \\
 &= \frac{1}{2\sqrt{\frac{x-1}{x+1}}} \cdot \frac{x+1-x+1}{(x+1)^2} \\
 &= \frac{\sqrt{x+1}}{2\sqrt{x-1}} \cdot \frac{2}{(x+1)^2} \\
 &= \frac{1}{\sqrt{x-1} \cdot \sqrt{(x+1)^3}} \\
 &= \frac{1}{\sqrt{(x-1)(x+1)^3}}
 \end{aligned}$$

References